Wi-Fi Training Program

Assignment Questions – Module 2

1. Brief about SplitMAC architecture and how it improves the AP's performance

Split-MAC architecture is a design principle used in centralized wireless networks where the functions of the 802.11 Media Access Control (MAC) layer are divided between two distinct entities: the **Lightweight Access Point (LAP)** and the **Wireless LAN Controller (WLC)**.

In traditional "Autonomous" or "Fat" APs, the AP handles every single task, from radio management to security and data switching. Split-MAC offloads the complex, "intelligence-heavy" tasks to the controller, leaving the AP to focus on time-sensitive, "real-time" radio functions.

How the MAC Layer is Split

1. Tasks Handled by the Lightweight AP (Real-time MAC)

The AP manages functions that require immediate, millisecond-level response times:

Frame Exchange: Sending and receiving data frames over the air.

Beacons and Probe Responses: Periodically announcing the network.

MAC Layer Encryption/Decryption: Handling the actual scrambling of data for security.

Real-time Signal Processing: Managing radio frequencies and power levels.

2. Tasks Handled by the WLC (Management MAC)

The Controller manages functions that are global or "intelligent" in nature:

802.11 Authentication and Association: Deciding if a device is allowed to join the network.

802.11i Key Management: Handling the complex handshake for WPA2/WPA3 security.

Radio Resource Management (RRM): Automatically adjusting the channels and power of all APs in the building to avoid interference.

Intrusion Detection: Monitoring for rogue access points or security threats.

Roaming Management: Orchestrating the "handoff" of a client from one AP to another.

How Split-MAC Improves AP Performance

Reduced Processor Overhead: By moving complex management and security handshakes to the WLC, the AP's onboard CPU is freed from heavy computational tasks. This allows the AP to dedicate 100% of its resources to high-speed data transmission and packet forwarding.

Centralized Intelligence: The AP no longer has to "think" about the entire network. The WLC has a bird's-eye view of all APs and can push configurations instantly. This prevents the AP from wasting cycles on suboptimal channel selections or management conflicts.

Seamless Roaming: In a Split-MAC environment, the WLC maintains the "state" of the client. When a user moves from AP1 to AP2, the new AP doesn't need to re-authenticate the user from scratch; the WLC simply updates the mapping. This results in faster transitions for the user.

Hardware Efficiency: LAPs can be manufactured with less expensive, specialized processors because they don't need to run a full management operating system. This makes large-scale deployments (hundreds of APs) more cost-effective.

Dynamic Adaptation: The WLC can detect a "hole" in coverage if one AP fails and instantly command neighboring APs to increase their power to fill the gap—a feat an autonomous AP cannot perform on its own.

2. Describe about CAPWAP, explain the flow between AP and Controller

CAPWAP (Control and Provisioning of Wireless Access Points) is a standard networking protocol that allows a central **Wireless LAN Controller (WLC)** to manage a collection of **Lightweight Access Points (LAPs)**.

It is the industry-standard evolution of the older LWAPP protocol. CAPWAP enables the **Split-MAC architecture** by providing the communication tunnel through which management and data traffic flow.

The Two CAPWAP Tunnels

When an AP connects to a controller, it establishes two distinct "tunnels" using **UDP**:

CAPWAP Control Channel (UDP Port 5246): * Used for management traffic (configuring the AP, firmware updates, and monitoring).

a. This traffic is **encrypted** by default (using DTLS) to ensure secure management.

CAPWAP Data Channel (UDP Port 5247): * Used for actual client data traffic.

b. Wireless frames from clients are encapsulated inside CAPWAP packets and sent to the WLC to be switched onto the wired network.
c. This channel is generally not encrypted by default to maintain high performance, though DTLS can be enabled.

The CAPWAP Flow: How an AP Joins a Controller

The process of an AP finding and connecting to a WLC is known as the **Discovery and Join Process**. It follows these specific steps:

Step 1: Discovery

The AP sends out a **CAPWAP Discovery Request**. It tries to find a controller using three main methods:

Broadcast: Sending a discovery message on its local subnet.

DHCP Option 43: The DHCP server provides the AP with the IP address of the WLC.

DNS: The AP looks for a specific hostname (e.g., CISCO-CAPWAP-CONTROLLER.local).

Step 2: Discovery Response

All WLCs that receive the discovery request respond with a **Discovery Response**, informing the AP of their current load and capacity.

Step 3: Join Request

The AP selects the best WLC (based on priority or current load) and sends a **CAPWAP Join Request**.

Step 4: Join Response & Image Download

The WLC sends a **Join Response**. At this stage, the WLC checks the AP's firmware version. If the versions don't match, the AP downloads the correct firmware image from the WLC and reboots.

Step 5: Configuration Update

Once the AP has the correct firmware and is joined, the WLC pushes the configuration (SSIDs, security settings, radio channels) to the AP via **Configure Request** and **Response** messages.

Step 6: Maintenance (Keepalive)

The tunnel is kept active using "Keepalive" messages (Echo Requests). If the AP stops receiving Echo Responses from the WLC, it assumes the controller is down and attempts to find a new one.

Why use CAPWAP?

Centralized Management: Change a password once on the WLC, and it applies to 1,000 APs instantly.

Layer 3 Roaming: Since all traffic is tunneled back to the WLC, a user can move across different subnets without their IP address changing.

Security: Management traffic is hidden from hackers on the local wire via DTLS encryption.

3. Where this CAPWAP fits in OSI model, what are the two tunnels in CAPWAP and its purpose

1. CAPWAP in the OSI Model

CAPWAP is considered a **Layer 3 (Network Layer)** protocol because it relies on IP addresses to establish communication between the AP and the Controller. However, in terms of its operation, it encapsulates higher-level data.

Transport Layer (Layer 4): CAPWAP uses **UDP** (User Datagram Protocol) as its transport mechanism.

Logical Placement: Since CAPWAP provides a "tunnel" for other protocols (like 802.11 wireless frames) and handles management tasks, it technically spans from the **Network Layer (L3)** up through the **Application Layer (L7)**, but it is fundamentally an **IP-based (L3) tunneling protocol**.

2. The Two CAPWAP Tunnels
A CAPWAP session actually consists of two separate logical "tunnels" running over different UDP ports.
A. The Control Tunnel (UDP Port 5246)
Purpose: This tunnel is the "management lane." It is used for administrative tasks between the WLC and the AP.
What flows here: * Configuration updates (pushing SSIDs, passwords).
- Firmware/Image downloads for the AP.
- Radio Resource Management (RRM) instructions.
- Status monitoring and "Keepalive" messages (Echo Requests).

Security: This channel is **always encrypted** using DTLS (Datagram Transport Layer Security) to prevent unauthorized changes to the network.
B. The Data Tunnel (UDP Port 5247)
Purpose: This tunnel is the "traffic lane." It carries the actual data generated by wireless clients (like your phone or laptop).
What flows here:
- Wireless 802.11 frames are "wrapped" (encapsulated) in CAPWAP headers at the AP.
- They are sent to the WLC, where they are "unwrapped" and placed onto the wired Ethernet network.

Security: By default, data traffic is **not encrypted** to ensure maximum speed. However, it can be encrypted if a specific DTLS license is applied.

4. Whats the difference between Lightweight APs and Cloud-based Aps

1. Lightweight APs (Controller-Based)
A Lightweight AP (LAP) is designed to work with a physical or virtual **Wireless LAN Controller (WLC)** located within the local private network (on-premises). It uses the **Split-MAC** architecture and **CAPWAP** tunnels to function.
Management: Locally managed via a WLC.
Traffic Flow: In most modes, client data is tunneled back to the WLC (Centralized Switching).
Internet Dependency: If the internet goes down, the APs continue to function perfectly because the "brain" (WLC) is still reachable on the local network.
Best For: Large campuses, hospitals, or high-security environments where data must stay within the private network.
Example: Cisco Aironet/Catalyst APs with a Cisco 9800 WLC.
Imagine a University campus. All 500 APs in the buildings are "Lightweight." They all build a CAPWAP tunnel to a massive controller sitting in the University's data center. The controller manages roaming as students walk between buildings.

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2. Cloud-Based APs

Cloud-based APs (also called Cloud-Managed) do not require a local physical controller. Instead, the management and control functions are hosted in the **public cloud** by the manufacturer.

Management: Managed via a web-based dashboard (browser or app).

Traffic Flow: Client data is usually switched locally at the AP (Local Switching). Only management and monitoring data are sent to the cloud.

Internet Dependency: If the internet goes down, the APs will continue to pass traffic for existing clients, but you cannot change settings, and new guest portal logins might fail until the connection to the cloud is restored.

Best For: Distributed businesses (retail chains with many small branches) or mid-sized offices that want "plug-and-play" simplicity without maintaining hardware controllers.

Example: Cisco Meraki or Aruba Instant On.

Imagine a coffee shop chain with 50 locations across the country. Each shop has one or two APs. The manager doesn't want to buy 50 controllers. They plug the Meraki AP into the internet; it "calls home" to the Meraki Cloud, downloads its config, and the manager controls all 50 shops from one website.

Detailed Comparison Table

Feature	**Lightweight AP**	**Cloud-Based AP**
Controller Location	Local (On-premises hardware/VM)	Public Cloud (Vendor's Data Center)
Communication Protocol	CAPWAP / LWAPP	Proprietary HTTPS/TLS-based
Data Path	Usually tunneled to WLC	Usually switched locally at the AP
Initial Cost	High (Must buy APs + Controller)	Low (Buy APs + Annual Subscription)
Scalability	Limited by Controller capacity	Virtually unlimited
Setup Complexity	High (Requires networking expertise)	Very Low (Zero-touch provisioning)
Subscription	Usually one-time hardware cost	Requires ongoing license/subscription

5. How the CAPWAP tunnel is maintained between AP and controller

1. The Echo Request and Response (Keepalives)
Once the CAPWAP tunnel is established, the AP and WLC exchange **Echo** messages at regular intervals to verify the path is still open.
Echo Request: The AP sends a CAPWAP Echo Request to the WLC.
Echo Response: The WLC must reply with a CAPWAP Echo Response.
Default Timer: Usually, this happens every **30 seconds**.
The Purpose: This confirms that the Layer 3 path (the internet or local network) is functioning and that the WLC process is still running.

2. Heartbeat Timers and Retries
If an Echo Response is not received, the system does not immediately drop the connection. It follows a specific "timeout" logic:
Echo Interval: The time between successful Echo Requests.
Echo Timeout: If a response is missed, the AP enters a "retry" state. It will send a few more requests in rapid succession.
Neighbor Dead Interval: If the AP fails to receive a response after the maximum number of retries (usually 3 to 5 attempts), it declares the WLC "dead."

3. The CAPWAP State Machine
The AP maintains its connection status through a state machine. To "maintain" the tunnel, the AP must stay in the **RUN State**.
Run State: In this state, the data and control tunnels are fully operational. The AP is providing Wi-Fi service to clients.
Transition on Failure: If the Heartbeat/Echo fails, the AP transitions from the **Run State** back to the **Discovery State**. It will immediately tear down its current SSIDs (in Local Mode) and start searching for a new controller to join.

4. DTLS Session Maintenance
Because the **Control Tunnel** is encrypted via **DTLS (Datagram Transport Layer Security)**, the tunnel maintenance also involves maintaining the security keys.
The AP and WLC periodically perform "re-keying" to ensure the encryption remains secure.
If the DTLS session expires or the keys become out of sync, the CAPWAP control tunnel will tear down and restart.

6. Whats the difference between Sniffer and monitor mode, use case for each mode
   1. Sniffer Mode
   In Sniffer Mode, a **Lightweight Access Point (LAP)** stops serving clients entirely and dedicates its radio to "sniffing" a specific channel. It captures every packet it hears on that channel and forwards them to a remote device (like a laptop running Wireshark) for analysis.

How it Works: The AP encapsulates the captured 802.11 wireless frames into **IP/UDP packets** and sends them over the wired network to the IP address of your analysis machine.
Key Behavior: The AP is **"Off-Channel"**—it cannot provide Wi-Fi access to any users while in this mode.
Use Cases:

- **Advanced Troubleshooting:** When you need to see exactly what is happening at the packet level (e.g., analyzing a failed 4-way security handshake).
- **Remote Analysis:** Capturing traffic from a remote branch office and viewing it on your screen at headquarters without physically being there.

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2. Monitor Mode

Monitor Mode is a dedicated state where the AP radio scans all available channels (channel hopping) to gather information about the environment. Unlike Sniffer Mode, it doesn't send full packet captures to a remote PC; instead, it reports statistics to the Wireless LAN Controller (WLC).
How it Works: The AP continuously cycles through all channels (e.g., 1, 2, 3... in 2.4GHz) for short periods. It listens for noise, interference, and unauthorized devices.
Key Behavior: Like Sniffer Mode, the AP **does not serve clients**. It acts purely as a sensor for the network.
Use Cases:

- **Rogue Detection:** Identifying "Rogue APs" (unauthorized routers plugged in by employees) or malicious "Evil Twin" attacks.
- **Intrusion Detection (WIDS/WIPS):** Detecting wireless attacks like deauthentication floods or MAC spoofing.
- **Location Services:** Helping to calculate the physical location of a client by measuring signal strength (RSSI) from multiple monitor-mode APs.

7. If WLC deployed in WAN, which AP mode is best for local network and how?
When the **Wireless LAN Controller (WLC)** is deployed across a Wide Area Network (WAN)—meaning the AP is at a branch office and the "brain" is at a central headquarters—the best AP mode to use is **FlexConnect Mode** (formerly known as H-REAP).

Why FlexConnect is the Best Choice

In a standard deployment (Local Mode), all user data traffic is tunneled through the WAN to the WLC. If the WLC is miles away, this creates massive latency and consumes expensive WAN bandwidth. **FlexConnect** solves this by allowing the AP to make its own switching decisions locally.

How FlexConnect Works (The Two States)

1. Connected Mode (WAN is UP)

When the link to the WLC is active, the AP maintains its CAPWAP control tunnel. However, you can configure the data traffic in two ways:

Local Switching: The AP drops user data directly onto the local branch switch. This keeps high-bandwidth traffic (like internal file transfers or video) off the WAN link.
Central Switching: Specific traffic (like a Guest SSID) can still be tunneled back to the WLC for centralized filtering or security.

2. Standalone Mode (WAN is DOWN)

This is the "survival" mode. If the WAN link fails and the AP loses its connection to the WLC:

The AP **continues to function** and broadcast the SSID.

It switches all traffic locally so users at the branch can still reach local printers, servers, and other colleagues.

Once the WAN link returns, the AP automatically rejoins the WLC and uploads any gathered statistics.

Key Performance Benefits

Bandwidth Conservation: By switching traffic locally, you don't waste WAN capacity sending local data to a remote headquarters just to have it sent back again.

Reduced Latency: Users get faster response times because their data doesn't have to travel to the HQ and back (the "hairpin" effect).

High Availability (Resiliency): If the internet/WAN goes down, the branch office doesn't lose Wi-Fi. In standard "Local Mode," the Wi-Fi would stop working entirely.

Local Authentication: FlexConnect can be configured to use a local RADIUS server or even "Local Auth" on the AP itself, ensuring users can still log in even if the HQ authentication server is unreachable.

8. What are challenges if deploying autonomous APs (more than 50) in large network like university

1. Scalability and Management Overhead

In an autonomous setup, every configuration change must be performed on each AP individually.

The Challenge: If you need to change a Wi-Fi password or add a new SSID across 50+ APs, an administrator must log into each device one by one.

The Impact: This is extremely time-consuming and prone to human error, leading to inconsistent security settings across the campus.

2. Lack of Centralized Radio Resource Management (RRM)

Autonomous APs do not coordinate their radio frequencies or power levels.

The Challenge: APs in close proximity may choose the same channel, leading to **Co-Channel Interference (CCI)**.

The Impact: In a university setting with high AP density (dormitories, lecture halls), the lack of automatic channel and power adjustment results in poor signal quality and "dead zones" that the APs cannot fix on their own.

3. Inefficient Roaming

Roaming is managed entirely by the client device in an autonomous network, with no assistance from the infrastructure.

The Challenge: Since APs don't share client "state" information, when a student walks between buildings, their device must perform a full re-authentication (including a new security handshake) with every new AP.
The Impact: This causes noticeable drops in connection, breaking video calls or active downloads during the transition.

4. Security Vulnerabilities

Security is difficult to enforce uniformly without a central controller.
The Challenge: If an AP is physically stolen, it contains the full network configuration and security keys in its local memory.
The Impact: Monitoring for **Rogue APs** (unauthorized routers) becomes nearly impossible, as there is no central system to "scan" the environment and alert administrators to threats.

5. No Load Balancing

Autonomous APs have no way to "steer" clients to less crowded neighbors.
The Challenge: If 100 students enter a single lecture hall, they will all try to connect to the nearest AP, even if an AP in the next room is idle.
The Impact: The overloaded AP will experience severe performance degradation, while the rest of the network's capacity remains underutilized.

6. Complex Firmware Updates

Keeping 50+ individual devices updated is a logistical nightmare.
The Challenge: Each AP must be manually updated with the latest firmware to patch security holes.
The Impact: If the process is skipped due to the sheer volume of work, the university network becomes vulnerable to exploits that have already been fixed by the manufacturer.

9. What happens on wireless client connected to Lightweight AP in local mode if WLC goes down.

In **Local Mode**, a Lightweight Access Point (LAP) is entirely dependent on the **Wireless LAN Controller (WLC)** for its "intelligence" and data forwarding. If the WLC goes down or the CAPWAP tunnel is interrupted, the following occurs:

Immediate Loss of Connectivity

Because Local Mode uses **Central Switching**, all user data must be encapsulated in a CAPWAP tunnel and sent to the WLC to be processed.
The Result: The moment the AP realizes the WLC is unreachable, it can no longer forward traffic. The "data plane" effectively breaks.

SSID Termination

The AP is programmed to stop providing service if it loses its connection to its controller.
The Result: The AP will **stop broadcasting the SSID** (the Wi-Fi network name will disappear).

Client Impact: Wireless clients (laptops, phones) will immediately lose their Wi-Fi signal and disconnect. They will likely begin searching for another available network or switch to cellular data.

AP Re-entry into Discovery Mode

Once the AP detects the WLC is down (via missed Echo Keepalives), it transitions out of the "Run State."

The Action: The AP enters the **Discovery State**. It will begin sending out CAPWAP Discovery Requests across the network, searching for a backup WLC or waiting for the primary WLC to come back online.

Visual Status: Often, the physical LED on the AP will change color (e.g., flashing red or green) to indicate it is in discovery mode and not currently serving clients.

CAPWAP State Machine Reset

The AP clears its association table. Even if the WLC comes back online a minute later, the clients must perform a completely new association and authentication process.

10.